

REQUEST FOR INFORMATION

1,024-GPU NVIDIA B300 AI Factory Infrastructure

Redacted Draft

Technical RFI for compute, control plane, monitoring, storage, networking, software, support and facility-facing requirements.

Document Attribute	Value
Document Type	Request for Information (RFI)
Solution Class	On-premises AI Factory
Target Scale	128 HGX B300 systems / 1,024 NVIDIA B300 GPUs
Response Type	Technical architecture, specifications, quantities, support and indicative facility requirements
Status	Reconstructed draft based on surviving supplier-response evidence and subsequent architecture reconciliation

Important status note. This document is a plausible reconstruction of the original RFI requirements. It is not represented as the authenticated historical source document. Requirements are phrased conservatively where the surviving response evidence does not prove whether a value was customer-mandated or supplier-selected.

1. Purpose and Background

The customer is seeking information and proposals from qualified infrastructure suppliers and integrators for an on-premises AI Factory based on NVIDIA B300 GPU systems. The target environment is 128 eight-GPU HGX B300-class systems, providing a total of 1,024 NVIDIA B300 GPUs, together with the supporting control-plane, monitoring, storage, network, software, out-of-band management and facility-facing infrastructure required for production operation.

The RFI is intended to establish a technically compliant architecture and indicative Bill of Materials / Bill of Quantities (BoM/BoQ), identify vendor options and support arrangements, and obtain the engineering information needed for subsequent detailed design, procurement and implementation planning.

2. Scope of Requested Response

- GPU compute systems and node-level connectivity.
- CPU-based cluster provisioning, Kubernetes/Slurm control-plane and administration infrastructure.
- Dedicated monitoring and observability infrastructure.
- High-performance shared storage capable of supporting AI training and inference workloads.
- GPU east-west/backend fabric, north-south/frontend fabric, storage, monitoring and control-plane networking.
- Physically separate out-of-band management network.
- Border / upstream data-centre connectivity.
- Software stack for orchestration, management, observability and data-centre monitoring.
- Optics, transceivers and cabling classes required by the proposed architecture.
- Indicative rack count, equipment density and IT-power requirements.
- Warranty, support, hardware replacement SLA and software subscription/licensing assumptions.

3. Bidder Response Instructions

For each requirement, the respondent should state the proposed manufacturer, model/part number, technical specification, quantity, unit of measure, compliance status, assumptions and any exception or dependency. Equivalent products may be proposed where the RFI is vendor-neutral, provided that the proposed configuration meets or exceeds the requested technical characteristics.

Response Field	Expected Bidder Content
Manufacturer / OEM	Proposed equipment or software vendor.
Model / Part Number	Exact commercial model or part number where available.
Technical Description	Configuration and key specifications.
Quantity	Total quantity required for the proposed architecture.
Compliance	Comply / Partially Comply / Exception.
Remarks / Assumptions	Dependencies, topology assumptions, alternative designs and clarification requests.
Support / Warranty	Term, hours of coverage and replacement SLA.
Commercial Information	Pricing and subscription terms where requested in the commercial response.

4. GPU Compute Infrastructure

4.1 Target Scale

Requirement	Target
GPU compute systems	128
GPU accelerator	NVIDIA B300
GPUs per system	8
Total GPUs	1,024
Platform class	HGX B300 NVL8-class or equivalent approved OEM implementation

4.2 Minimum / Indicative Node Specification

Component	Requested Characteristic
CPU	Dual-socket server CPU configuration, approximately 64 cores per socket and >=2.4 GHz class.
System Memory	Approximately 3 TB DDR5-class memory per GPU node.
OS Storage	2 x approximately 1.92 TB enterprise SSD, RAID-1 or equivalent resilient boot arrangement.
Local Data Storage	4 x approximately 7.68 TB enterprise NVMe SSD.
GPU Subsystem	8 x NVIDIA B300 GPUs on HGX B300 NVL8-class platform.
Scale-up Interconnect	NVIDIA NVLink / NVSwitch integrated within the HGX system.
East-West Interfaces	8 x high-speed 800G-class GPU fabric interfaces / SuperNICs per node.
North-South Interface	BlueField-3 DPU-class connectivity, dual-port 400GbE class preferred.
Power Supplies	Redundant hot-swap, high-efficiency / Titanium-class.
Cooling	Air-cooled server implementation unless bidder identifies a mandatory OEM alternative.
Support	3-year-class, 24x7 support with approximately 4-hour hardware replacement target.

Bidder clarification requested: Identify any deviation between the proposed BlueField-3 configuration and a dual-400GbE target, including bandwidth impact and whether an alternative DPU SKU is required.

5. CPU Control-Plane Infrastructure

The respondent shall propose CPU-only infrastructure for cluster provisioning, management, scheduler control and Kubernetes administration. The final functional split may be proposed by the bidder, but the architecture should provide redundant control services and avoid consuming B300 GPU capacity for routine platform administration.

Component	Indicative Requirement
CPU	Dual socket, >=32 cores/socket, >=2.0 GHz class.
Memory	Approximately 16 x 64 GB DDR5 or equivalent >=1 TB configuration.
OS Storage	2 x approximately 960 GB SSD, RAID-1.
Data / Service Storage	2 x approximately 7.68 TB enterprise SSD.
High-Speed Network	Dual 400GbE-class connectivity.
Management Network	Multiple 1GbE-class management interfaces.
Power / Cooling	Redundant hot-swap PSU; air cooling.
Support	3 years, 24x7, approximately 4-hour hardware replacement.

Expected functions include bare-metal / cluster provisioning, Slurm control, Kubernetes administrative control-plane services, platform services and associated management functions. The bidder should identify the recommended number of nodes by role and the high-availability model.

6. Monitoring Infrastructure

Provide a dedicated monitoring and observability infrastructure capable of collecting, storing, visualizing and alerting on infrastructure, GPU, network, storage, operating-system, Kubernetes and scheduler telemetry.

Role	Indicative Quantity	Requirement
Monitoring control-plane nodes	3	Redundant monitoring / management control services.
Monitoring worker / data nodes	3	Metrics, logs, dashboards, alerting and observability data services.
Total monitoring nodes	6	Production monitoring cluster.

Component	Indicative Requirement
CPU	Dual 32-core-class CPUs.
Memory	Approximately >=1 TB DDR5.
Boot	Resilient / RAID-1 SSD boot.
Local Data	Approximately 7.68 TB-class enterprise SSD/NVMe capacity as proposed.
Network	Dual high-speed Ethernet, up to 400GbE class where required by architecture.
Support	3-year, 24x7-class enterprise support.

Supply-chain requirement: Where open-brand servers are proposed for CPU or monitoring roles, the respondent should disclose OEM country of origin and comply with any customer restriction on Chinese

OEMs.

7. High-Performance Shared Storage

The respondent shall propose a scale-out shared storage platform suitable for large AI training, inference and data-ingestion workloads. The solution should support a single namespace or equivalent operational model and provide resilient high-bandwidth access from the compute estate.

Metric	Requested Target
Usable capacity	4 PB
Aggregate read throughput	250 GB/s
Aggregate write throughput	100 GB/s
Read IOPS	Approximately 900,000 to 1,000,000
Write IOPS	Approximately 300,000
Management / control	Redundant management capability; bidder to identify whether dedicated management servers are required.
Support	24x7 enterprise support with approximately 4-hour replacement target.

Performance interpretation: The performance figures should be treated as aggregate storage-system objectives unless the customer explicitly confirms a different scope. Bidder shall state benchmark assumptions including block size, concurrency, protocol and measurement method.

The response should include storage-node count, drive count and capacity, raw and usable capacity, protection / failure-domain assumptions, network interfaces per backend, expected aggregate throughput and IOPS, expansion method, supported protocols and any GPU Direct Storage or equivalent acceleration.

8. Networking Requirements

The respondent shall provide an end-to-end network architecture covering the GPU backend/east-west fabric, frontend/north-south access, CPU control-plane, storage, monitoring, out-of-band management and upstream/border connectivity. All proposed switch counts, optics and interconnect quantities shall be explained by topology and port-count calculations.

8.1 GPU Backend / East-West Fabric

- Provide a high-bandwidth, low-latency fabric for all 128 HGX B300 systems.
- Provide eight high-speed fabric attachments per B300 node where required by the selected B300 platform/reference design.
- Design for non-blocking or otherwise explicitly stated oversubscription, with redundant topology and clear failure domains.
- Provide leaf/spine or equivalent topology, switch models, port utilization, optics and cabling quantities.

- State whether the proposed fabric uses InfiniBand or Ethernet and justify the choice for distributed GPU workloads.

8.2 Frontend / North-South Network

- Provide redundant north-south connectivity for user, platform, service and external traffic.
- Support dual-attached B300/DPU connectivity and high-bandwidth service access.
- Provide a leaf/spine Clos or equivalent scalable architecture and identify blocking ratio.
- Include connectivity for storage where the bidder proposes a physically converged Ethernet fabric, while preserving logical isolation.

8.3 CPU, Storage and Monitoring Networks

Provide redundant connectivity for control-plane servers, storage backends and monitoring systems. The bidder may propose dedicated fabrics or a converged design where performance, fault isolation and security requirements are demonstrably met.

8.4 Out-of-Band Management Network

- Provide a physically separate OOB network for BMC/IPMI/Redfish interfaces, switch management, storage management and other infrastructure devices.
- Provide sufficient spare-port headroom; a design that consumes 100% of available access ports is not acceptable without an approved expansion mechanism.
- Provide redundant uplinks and a controlled administrative path through enterprise identity, MFA, PAM/bastion or equivalent controls.

8.5 Border / External Connectivity

Provide redundant border connectivity from the AI Factory frontend network to the customer data-centre core, WAN and/or external service environment. The bidder shall state the assumed external bandwidth and identify any dependency on customer-provided upstream connectivity, firewall throughput or DDoS services.

9. Software and Platform Requirements

Capability	Requirement / Bidder Response
Container Orchestration	Kubernetes-based enterprise platform. Vanilla/upstream Kubernetes or equivalent may be proposed; bidder shall identify support model.
Kubernetes Management	Enterprise / next-generation Kubernetes fleet, policy and lifecycle management platform.
HPC / Batch Scheduling	Slurm or equivalent supported scheduler for distributed training and HPC/batch workloads.
GPU Software	NVIDIA drivers, CUDA, NCCL, container toolkit and required GPU/network operators.
Monitoring / Metrics	Prometheus-compatible collection, dashboards and alerting or equivalent.
GPU Telemetry	NVIDIA DCGM / DCGM Exporter or equivalent supported telemetry.
Centralized Logging	Elastic/OpenSearch-class or equivalent enterprise logging platform.
Observability	Integrated metrics, logs, alerting and operational dashboards.
Security / Audit	Centralized audit/event forwarding and SIEM integration capability.
DCIM / Facility Monitoring	Data-centre infrastructure monitoring / DCIM capability, including licensing model.

Capability	Requirement / Bidder Response
Storage Integration	CSI and other orchestration integration required by the selected storage platform.

10. Monitoring and Management Architecture

The bidder shall describe the high-level architecture by which the AI Factory is operated, monitored, alerted and audited. The monitoring and management system should provide visibility from physical infrastructure through compute, network and storage to platform and workload layers.

- HGX B300 node and GPU health, utilization, thermals, memory, power and fault telemetry.
- Backend/east-west and frontend/storage switch health, port state, errors, utilization and congestion telemetry.
- Storage node, capacity, throughput, latency, protection and service health.
- Kubernetes cluster, node, pod and control-plane monitoring.
- Slurm scheduler and job monitoring.
- Centralized logging, audit records, alerting and incident integration.
- Integration with enterprise ITSM and SIEM/SOC where available.
- Clear separation between cloud/SaaS management components and on-premises managed components, including secure control-channel direction.

11. Warranty, Support and Service Requirements

Area	Requested Response
Hardware warranty	State warranty term for all server, storage and network equipment.
Coverage	24x7 enterprise support preferred.
Replacement SLA	Approximately 4-hour hardware replacement target where commercially available.
Software support	State support/subscription term and entitlement for all commercial software.
Escalation	Provide support escalation model and named service organization.
Spares	Identify recommended on-site or regional spares for critical components.
Firmware / software lifecycle	Describe patch, firmware, driver and compatibility lifecycle support.

12. Rack, Power and Facility-Facing Information

The respondent shall provide sufficient equipment and facility-facing information to support downstream data-centre engineering. The response is not expected to replace detailed electrical, mechanical or structural design.

Information Requested	Bidder Response
Indicative rack count	GPU, network, storage/control/monitoring and total racks.
Rack population assumption	Systems per rack and any dedicated network/storage rack assumptions.

Information Requested	Bidder Response
IT power	Indicative maximum/typical IT load by equipment class and total.
Cooling	Air/liquid cooling requirements and OEM environmental constraints.
A/B power	Redundancy expectations and PSU feeds.
Physical dimensions / weight	Equipment dimensions and weight required for rack/floor planning.
Management interfaces	PDU/CDU/BMS/EPMS interfaces required by the IT solution.
Dependencies	Any vendor site-readiness or deployment prerequisites.

Scope boundary: Detailed rack elevations, breaker/PDU/busway engineering, CDU/hydraulic design, facility cooling plant, structural loading, UPS/generator capacity and site integration are downstream facilities-engineering activities.

13. Required Bidder Deliverables

1. Completed requirement-response matrix.
2. High-level solution architecture diagram.
3. Network topology diagrams for backend/east-west, frontend/north-south, storage and OOB fabrics.
4. Storage architecture and performance/capacity sizing basis.
5. Detailed BoM/BoQ by equipment class, including optics/transceivers and required software subscriptions.
6. Support and warranty schedule.
7. Rack and IT-power summary.
8. List of assumptions, exclusions, dependencies and technical deviations.
9. Indicative delivery lead times and implementation dependencies.
10. Identification of any customer decisions required before final quotation or detailed design.

14. Technical Response Matrix

The bidder should complete a line-item response substantially equivalent to the following structure. Additional technical detail may be provided in appendices.

#	Domain	Requirement	Proposed Product	Qty	Compliance	Remarks
1	GPU Compute	128 HGX B300-class systems / 1,024 GPUs				
2	GPU E-W Fabric	High-speed backend fabric for all B300 nodes				
3	N-S / Frontend	Redundant high-speed frontend fabric				
4	Control Plane	CPU-based provisioning, K8s and Slurm control services				
5	Monitoring	6-node-class monitoring/observability platform				

#	Domain	Requirement	Proposed Product	Qty	Compliance	Remarks
6	Storage	4 PB usable; 250 GB/s read; 100 GB/s write; target IOPS				
7	OOB	Physically separate management network with headroom				
8	Border	Redundant upstream/core connectivity				
9	Software	Kubernetes, management, Slurm, observability, DCIM				
10	Support	3-year-class 24x7 support / ~4-hour replacement				